Journal of Alloys and Compounds 817 (2020) 152731

Contents lists available at ScienceDirect

Journal of Alloys and Compounds

journal homepage: http://www.elsevier.com/locate/jalcom

Effects of microstructure of Ni barrier on bonding interface diffusion behaviors of Bi–Te-based thermoelectric material

Ekubaru Yusufu [a, *], Tohru Sugahara [a], Michio Okajima [b], Shutaro Nambu [b], Katsuaki Suganuma [a]

[a] Department of Advanced Interconnection Materials, The Institute of Scientific and Industrial Research, Osaka University, Mihogaoka 8-1, Ibaraki, Osaka, 567-0047, Japan
[b] E-ThermoGentek Co., Ltd., 102 Kujyo CID bld., 13, Shimotonodacho, Higashikujo, Minami-ku, Kyoto, 601-8047, Japan

ARTICLE INFO

Article history:
Received 31 August 2019
Received in revised form
18 October 2019
Accepted 19 October 2019
Available online 22 October 2019

Keywords:
Thermoelectric module
Ag paste
BiTe
Diffusion barrier
Bonding technology

ABSTRACT

To enhance the conversion efficiency of thermoelectric generation (TEG) devices, they must withstand larger temperature differences. Previously, a large-scale flexible TEG device was fabricated for use in energy harvesting. However, its maximum operating temperature was limited to approximately 150 °C owing to the low melting point of the Sn–Ag-based solder (~150 °C) that bonds TE materials and electrodes. Therefore, the present study attempted to develop a bonding interface resistant to heat up to 250 °C. Ag paste was chosen as the bonding material instead of the Sn–Ag-based solder because it has many advantages such as high-temperature stability (melting point ~ 960 °C), printability, and low electrical resistivity. Ni/Au was used as a diffusion barrier layer. Ni/Au layers were prepared by sputtering and electroplating, respectively. Investigation of the element diffusion behaviors of these two bonding interfaces at 250 °C showed that electroplated Ni blocked electrode diffusion, but Ni itself significantly diffused into n-type Bi_2Te_3 and formed a NiTe phase. Conversely, sputtered Ni could not block electrode diffusion, but Ni hardly diffused into n-type Bi_2Te_3. This work provides a fundamental understanding of the grain-boundary diffusion and diffusion properties of crystals and amorphous materials.

© 2019 Elsevier B.V. All rights reserved.

1. Introduction

Thermoelectric (TE) devices can generate electricity from a temperature difference according to the Seebeck effect. Conversely, they can produce a temperature difference with the application of an electrical current according to the Peltier effect [1–5]. Thus, TE devices can be used for both power generation and solid-state refrigeration [4,5]. TE devices mainly consist of p- and n-type semiconducting TE materials, metal electrodes, and an insulating substrate. Generally, TE generation (TEG) devices are fabricated by bonding p- and n-type TE materials in a π pattern with metal electrodes and sandwiching them between two insulating substrates with high thermal conductivity. Heat flows from the bottom substrate to the top, making all thermoelectric elements thermally in parallel and electrically in series. TE devices have many advantages, such as the ability to recover waste heat over a very wide temperature range without any moving parts, noise, gas emission, and maintenance. Owing to such advantages, TE devices have been widely applied to various fields such as remote power supply (radioisotope thermoelectric generators), automobiles, temperature sensors, control devices, water condensation, and implantable/wearable devices [3,4,6,7].

The effectiveness of a TE device is represented by the dimensionless figure of merit, $ZT = (S^2T)/RK$, where S, R, K, and T are the total Seebeck coefficient, total electrical resistance, total thermal conductance, and absolute temperature, respectively [3]. Here, R and K can also be expressed as $R = R_c + R_m$ and $K = K_c + K_m$, respectively, where R_m and K_m are the electrical and thermal resistance of the material and R_c and K_c are the electrical and thermal resistance of the contact, respectively. This indicates that the performance of a TE device depends not only on the intrinsic TE material properties but also the thermal and electrical contact resistance (R_c and K_c), which are additional important variables that should be considered to improve the ZT and reliability of the device [8–12]. Therefore, the performance and reliability of TE devices are severely limited by the interfacial performance of the bonding between the TE materials and electrodes.

* Corresponding author. Tel.: +81 6 6879 8521; fax: +81 6 6879 8522.
E-mail address: ekubaru@eco.sanken.osaka-u.ac.jp (E. Yusufu).

https://doi.org/10.1016/j.jallcom.2019.152731
0925-8388/© 2019 Elsevier B.V. All rights reserved.

The interfacial properties include the adhesiveness in the interface, metal–semiconductor (M–S) contact resistance, and blocking of the interfacial element diffusion between the TE materials and electrodes. To block element diffusion, it is necessary to set a diffusion barrier layer in the bonding interface, and diffusion barriers significantly influence the reliability and stability of TEG devices. It is known that element diffusion is mainly divided into crystal-lattice diffusion, which includes interstitial diffusion, vacancy diffusion, and substitution diffusion; grain-boundary diffusion; dislocation diffusion; and surface diffusion. Crystal-lattice diffusion generally occurs at high temperature because atoms on the lattice node are relatively stable. However, atoms near a grain boundary or dislocation have higher energy and easily diffuse along these defects [13,14]. From this knowledge, it can be understood that the reduction of grain boundaries and increase of crystallinity of the diffusion barrier are effective ways to increase the blocking performance of the diffusion barrier. It was reported that some refractory metals, such as W, Ti, Zr, and their nitrides or alloys, are excellent barriers to block atomic diffusion because of their high melting point and chemical stability. Thus, the challenge to optimize interfacial adhesion and control element diffusion is to choose an appropriate diffusion barrier layer [15–17].

Several studies have been reported on Bi–Te-based TE devices, for which Ni is always used as the diffusion barrier [18–21]. Recently, our group developed a Bi–Te-based TEG device by using conventional semiconductor packaging technology [22]. In that device, Ni/Au was used as the diffusion barrier, and Sn–Ag-based solder was used as the bonding material. Although Bi–Te-based TE materials can be used up to approximately 300 °C [23,24], the maximum operating temperature of the device was limited to approximately 150 °C owing to the low melting point of the Sn–Ag-based solder (~150 °C) that bonds Bi–Te-based TE materials and Cu electrodes [22]. Therefore, in order to apply higher temperature differences to the device, in this study, we attempted to develop a bonding interface resistant to heat up to 250 °C. Ni/Au was retained as the diffusion barrier layer, but Ag paste was chosen as the bonding material, instead of the Sn–Ag-based solder, because it has many advantages such as high-temperature stability (melting point ~ 960 °C), printability, low electrical resistivity, high thermal conductivity (150 $Wm^{-1}K^{-1}$), and a temperature requirement of less than 300 °C for sintering-bonding [25–28]. The Ni/Au diffusion barrier layer was prepared using two different deposition processes, sputtering and electroplating, and the element diffusion behaviors of these two bonding interfaces at 250 °C were investigated.

2. Results and discussion

Fig. 1 shows cross-section scanning electron microscopy (SEM) images of the experimental TEG samples in which the Ni/Au layer was sputtered or electroplated before and after heat-aging Bi–Te chips at 250 °C for 100 h. In the chip on which Ni/Au was sputtered before heat aging, the grain boundaries of the Ni layer can be clearly observed, and the grains grew in a columnar structure with voids at the grain boundary, as shown in Fig. 1 (a). In contrast, for the chip on which Ni/Au was electroplated before heat aging, the Ni layer was dense without any voids; however, no grain boundaries were observed, which implies that the grain size of the electroplated Ni layer is much less than that of the sputtered Ni layer, as shown in Fig. 1 (b). On the other hand, after heat aging at 250 °C for 100 h, slight grain growth occurred in the sputtered Ni layer, but almost no diffusion of Ni was observed, as shown in Fig. 1(c). However, an extra phase was observed near the Ni/BiTe interface in the electroplated chip after heat aging, and X-ray diffraction (XRD) and energy-dispersive X-ray (EDX) analyses revealed that the extra phase was NiTe, as shown in Fig. 1 (d), which is consistent with a previously reported study [19].

Fig. 2 shows the bulk-surface XRD pattern of sputtered and electroplated chips before and after heat aging, obtained by controlling the incident depth of X-rays. As shown in Fig. 2, all the XRD peaks of the chips sputtered and electroplated before heat aging are broad, implying that the grain sizes of the Au layer were very small, and these peaks corresponded to Bi_2Te_3 (JCPDS Card No. 00-015-0863), Ni (JCPDS Card No. 00-004-0850), and Au (JCPDS Card No. 00-004-0784). However, for the chips sputtered and electroplated after heat aging, the XRD peaks of the sputtered chip are slightly sharper than that of the electroplated chip, implying that grain growth occurred during heat aging. However, extra XRD peaks appeared in the XRD patterns of the electroplated chip, and these peaks corresponded to NiTe (JCPDS Card No. 01-089-2679). The SEM and XRD results establish that, during heat aging, electroplated Ni diffused into the Bi_2Te_3 chip and formed a NiTe phase, whereas sputtered Ni showed almost no diffusion.

The transmission electron microscopy (TEM) microstructure and selected area electron diffraction (SAED) pattern of the Ni sputtered and electroplated layer before heat aging are shown in Fig. 3 (a) - (e), respectively. The comparison of SAED pattern of Ni sputtered and electroplated as shown in Fig. 3 (c) and (d), respectively. As shown in Fig. 3 (c), the SAED of sputtered Ni was almost bright spots. Whereas, as shown in Fig. 3 (d), the ring pattern including bright spots was clearly observed in the SAED of electroplated Ni. These

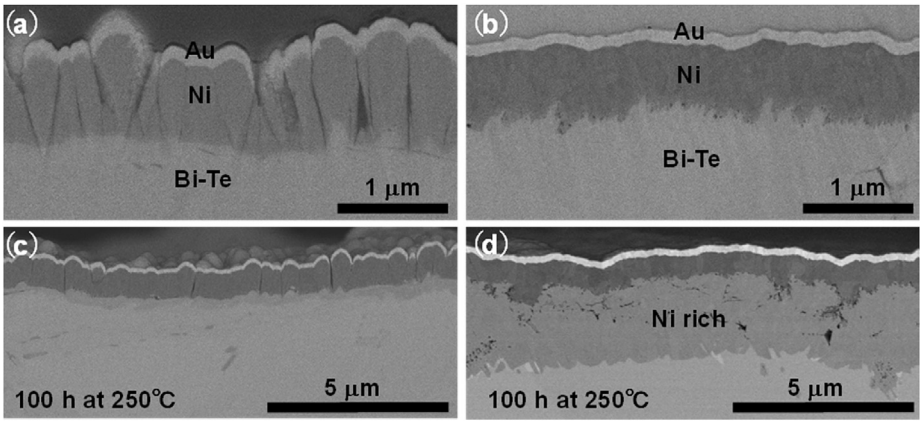


Fig. 1. Cross-sectional SEM microstructure images of Ni/Au sputtered (a) before and (c) after heat aging Bi–Te chips and Ni/Au electroplated (b) before and (d) after heat aging Bi–Te chips.

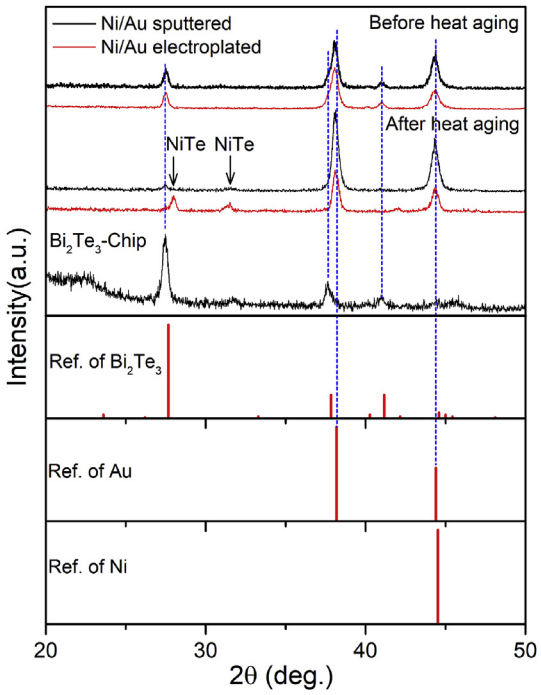


Fig. 2. Bulk-surface XRD patterns of Bi—Te chips with Ni/Au sputtered and electroplated before and after heat aging at 250 °C for 100 h along with reference data, obtained by controlling the incident depth of X-rays.

behaviors indicate that the crystallinity of sputtered Ni higher than that of electroplated Ni as well as the existence of amorphous phases in the electroplated Ni. This was also confirmed by high magnification TEM image, as shown in Fig. 3 (e), sputtered Ni was very highly crystalline with almost no amorphous phase, in contrast, the grain size of electroplated Ni is only several nanometers, and it includes many amorphous Ni phases. Therefore, it is considered that the sputtered Ni almost did not diffuse into the Bi—Te chips because of the stability of the crystalline phase, and the electroplated Ni strongly diffused into Bi—Te chips because of the instability of the nanocrystalline—amorphous phase.

Fig. 4 shows cross-sectional SEM images of Ni/Au sputtered/electroplated Bi—Te chips bonded with Ni/Au sputtered/electroplated Cu substrates by using sintered Ag paste before and after heat aging at 250 °C for 100 h. Before heat aging, in the Ni/Au sputtered sample, the interface was seeming very well formed without cracks, pores, or extra phases, in contrast, in the Ni/Au electroplated sample, the NiTe phase appeared in the Bi—Te chip near the bonded interface, as shown in Fig. 4 (a) and (b). On the other hand, after heat aging at 250 °C for 100 h, in the Ni/Au sputtered sample, many cracks and extra phases appeared in the Bi—Te chip, however, there was no difference between the samples electroplated before and after heat aging, as shown in Fig. 4 (c) and (d). EDX mapping (see supporting information) and line analysis were performed to determine the element distribution of the bonding interface. Here, only the element line profiles are shown.

Fig. 5 (a) and (b) shows the element line profiles of the sputtered bonding interface before and after heat aging, respectively, at

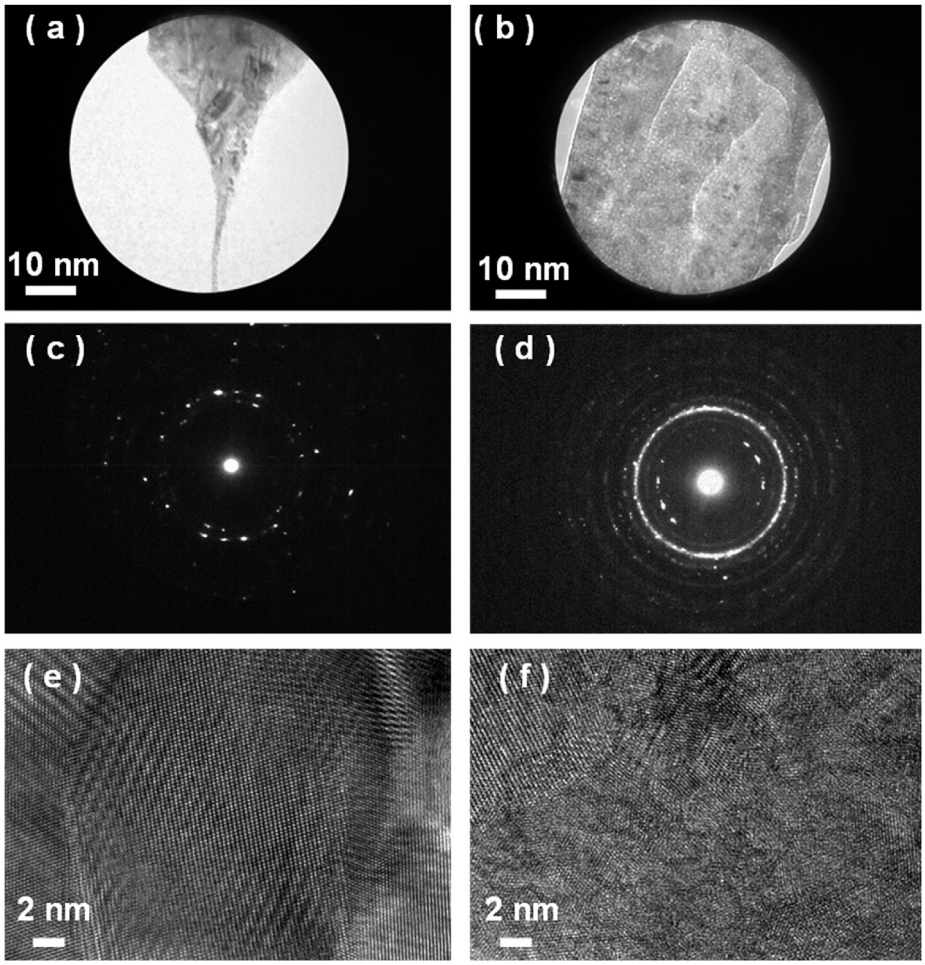


Fig. 3. TEM, SAED, and HRTEM images of (a), (c), and (e) sputtered and (b), (d), and (f) electroplated Ni layer before heat aging at 250 °C for 100 h.

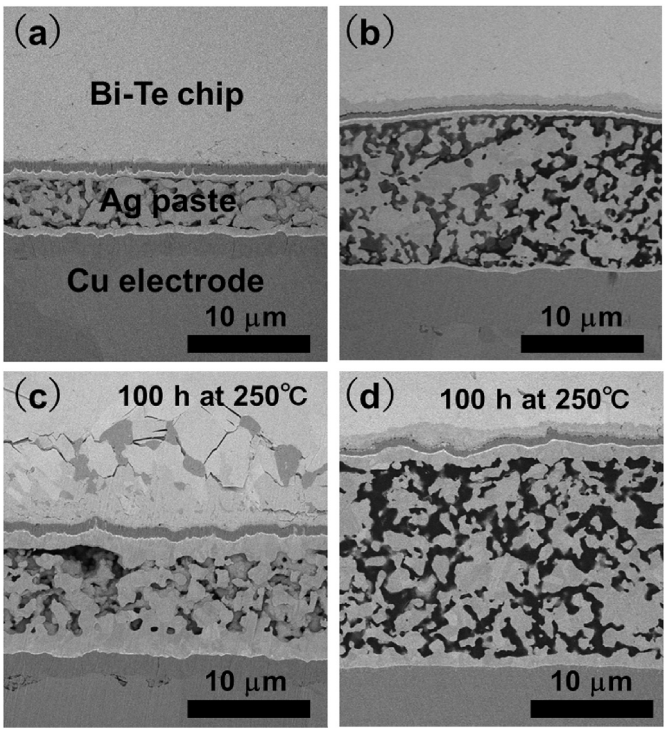


Fig. 4. Cross-sectional SEM microstructure images of Ni/Au sputtered (a) before and (c) after heat aging and Ni/Au electroplated (b) before and (d) after heat aging Bi–Te chips bonded with Ni/Au sputtered/electroplated Cu substrates by using sintered Ag paste.

250 °C for 100 h, where the element distribution is clearly shown in the cross-sectional view. As shown in Fig. 5 (a), before heat aging, the intensity of Cu in the Ag paste at the position marked by the red oblong indicates that Cu atoms have diffused into the Ag paste during the fabrication process. After heat aging, as shown in Fig. 5 (b), the intensities of Cu and Te in the Bi–Te chip at the positions marked by the red block and oblong, respectively, indicate that Cu atoms have strongly diffused into Bi–Te chips and formed a Cu-, Te–rich and Bi–poor phase. Furthermore, the O intensity in the Bi–Te chip at the position marked by the red oblong shows that O atoms have also diffused into the Bi–Te chips and make the chips oxidize near the interface area during the heat aging.

Fig. 6 (a) and (b) shows the element line profiles of the electroplated bonding interface before and after heat aging at 250 °C for 100 h. As shown in Fig. 6 (a), before heat aging, the intensities of Ni and Te in the bonding interface at the position marked by the red oblong indicate that Ni atoms have diffused into Bi–Te and formed a Ni–Te-rich phase; moreover, Ni atoms diffuse into the Cu electrode during the fabrication process. After heat aging, as shown in Fig. 6 (b), the intensities of Ni and Te are almost the same as those before heat aging. However, O peaks additionally appeared in Ni–Te phase, which indicates that the Ni–Te phase may be oxidized during the heat aging.

3. Conclusions

In this work, we developed n-type Bi_2Te_3 chips, with a sputtered or electroplated Ni/Au barrier layer, that were bonded with a Cu substrate with a sputtered or electroplated Ni/Au layer, and the reliability of the chips at 250 °C was investigated. The contributions of this study can be summarized as follows.

SEM microstructure analysis revealed that sputtered Ni layers grew in columnar shapes with voids at grain boundaries. On the other hand, electroplated Ni layers were very dense and without any voids. TEM microstructure analysis revealed that the sputtered Ni crystal grains had a size of several tens of nanometers and were very highly crystalline and orientational. On the other hand, the electroplated Ni crystal grains had a size of only a few nanometers and showed lower crystallinity with the presence of amorphous phases. These results imply that the sputtered Ni was in a stable phase because of its high crystallinity; however, the electroplated Ni was unstable because of its lower crystallinity. The electroplated Ni was significantly diffused into the Bi_2Te_3 chip to form a NiTe phase, but

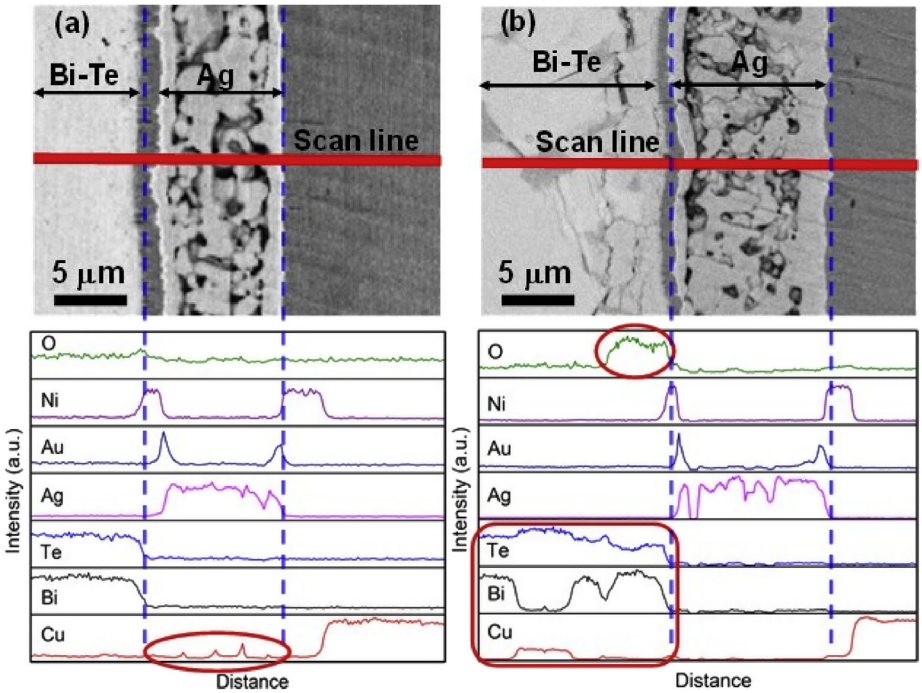


Fig. 5. Cross-sectional SEM images and EDX element line profiles of Ni/Au sputtered bonding interface (a) before and (b) after heat aging at 250 °C for 100 h.

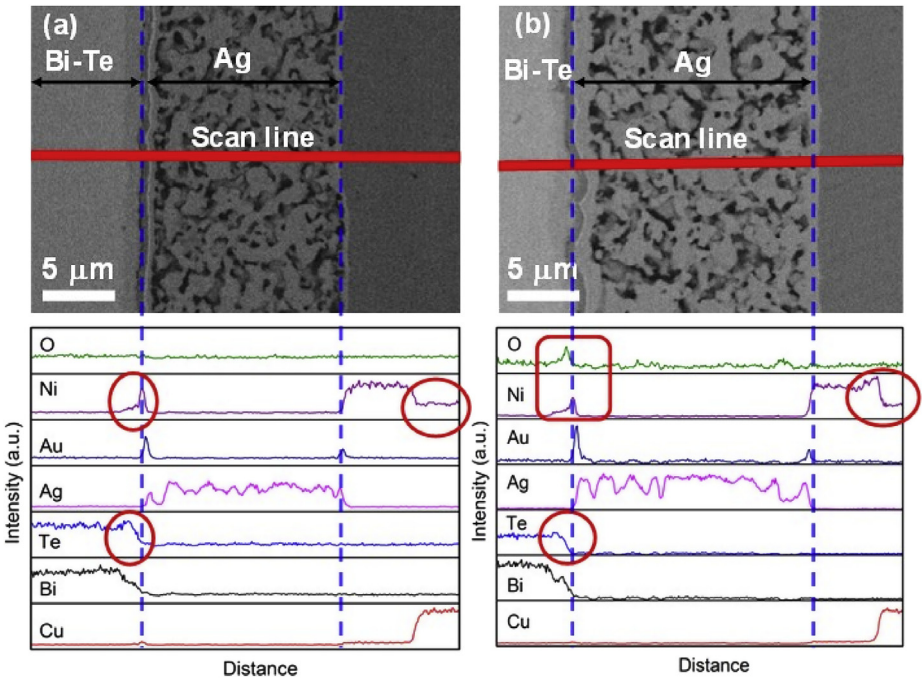


Fig. 6. Cross-sectional SEM images and EDX element line profiles of Ni/Au electroplated bonding interface (a) before and (b) after heat aging at 250 °C for 100 h.

the sputtered Ni was hardly diffused. In the bonded sample with sputtered Ni/Au, the Cu electrode diffused violently into the chip and formed a Cu–Te-rich phase. On the other hand, in the bonded sample with electroplated Ni/Au, the diffusion of Cu was blocked, but a NiTe phase formed on the chip side. The above results demonstrate that, with sputtered Ni/Au, grain-boundary diffusion is predominant and Cu electrodes diffused along the grain boundaries, while with electroplated Ni/Au, self-diffusion is predominant because of the instability of the monocrystalline–amorphous phase. These results suggest that the highly crystalline Ni layer, which has dense grain boundaries and no pores, is a highly effective diffusion barrier. Based on the above results, it was considered that the bonding combination of sputtered and electroplated process such as the Ni/Au electroplated Cu electrode and Ni/Au sputtered Bi–Te based chip is advantageous to blocking the element diffusions.

4. Experimental section

As shown in Fig. 7, 1-μm-thick Ni and 0.1-μm-thick Au sputtered/electroplated on $30 \times 30 \times 1$ mm Bi–Te chips were introduced for bonding with 5-μm thick Ni and 0.1-μm-thick Au sputtered/electroplated on a $100 \times 10 \times 0.8$ mm Cu substrate. A 5 μm Ni barrier layer was deposited using RF sputtering at power 100 W on both the chip and the substrate. A 0.1 μm Au layer was then deposited using DC sputtering. The base pressure of the vacuum chamber was $<5.0 \times 10^{-3}$ Pa, and the Ar flow rate was 10 sccm. The substrate was maintained at room temperature during deposition. The electroplating was done by Kiyokawa Electronics Co., Ltd. Here, Au acted as an adhesive for bonding with the use of Ag paste. Firstly, Ag paste was printed on the Cu substrate, following which the Bi–Te chips were mounted on the Ag paste. Subsequently, this structure was placed on a hot plate and sintered at 250 °C for 1 h in air. The heat-aging process was conducted at 250 °C for 100 h in air. The X-ray diffraction (XRD) patterns of the bulk samples were collected in the range of $2\theta = 5 - 50°$ using an X-ray diffractometer (Rigaku SmartLab, Akishima, Tokyo, Japan) with Cu K α radiation at room temperature at 45 kV and 100 mA, using a step size of 0.02°, scanning rate of 4°/min, and incidence angle, ω, of 0.2°. The cross section of bonds was prepared using an ion-milling machine (IM4000, Hitachi, Tokyo, Japan), and microstructure analysis was performed using field-emission SEM and EDX analysis (SU8020, Hitachi, Tokyo, Japan). The cross section of sinter Ag bonding was prepared using the ion-milling machine (IM4000, Hitachi, Japan), and the microstructure was observed by field-emission SEM (SU8020, Hitachi, Tokyo, Japan). TEM (JEM-ARM200F, JEOL) was employed to investigate the sputtered/electroplated Ni microstructure at 200 kV.

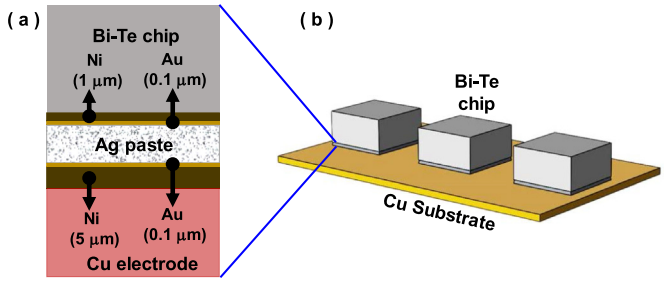


Fig. 7. Schematic of the experimental sample: (a) structure of the bonding interface; (b) sample after the bonding of the chip and substrate.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was partially supported by the Program for Advancing Strategic International Networks to Accelerate the Circulation of Talented Researchers from Japan Society for the Promotion of

Science (JSPS) and the Management Expenses Grant for National Universities Corporations from the Ministry of Education, Culture, Sports, Science and Technology of Japan (MEXT).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jallcom.2019.152731.

References

[1] H.J. Goldsmid, Thermoelectric Refrigeration, Plenum Press, New York, 1964.
[2] G.G. Yadav, J.A. Susoreny, G. Zhang, H. Yang, Y. Wu, Nanostructure-based thermoelectric conversion: an insight into the feasibility and sustainability for large-scale deployment, Nanoscale 3 (2011) 3555.
[3] G.J. Snyder, E.S. Toberer, Complex thermoelectric materials, Nat. Mater. 7 (2008) 105.
[4] J. Yang, T. Caillat, Thermoelectric materials for space and automotive power generation, MRS Bull. 31 (2006) 224.
[5] L.E. Bell, Cooling, heating, generating power, and recovering waste heat with thermoelectric systems, Science 321 (2008) 1457–1461.
[6] Jiabin Yan, Xiaoping Liao, Deyang Yan, Y. Chen, Review of micro thermoelectric generator, J. Microelectromech. Syst. 27 (2018) 1–18.
[7] C. Yu, K.T. Chau, Thermoelectric automotive waste heat energy recovery using maximum power point tracking, Energy Convers. Manag. 50 (2009) 1506–1512.
[8] H.J. Goldsmid, in: D. Rowe (Ed.), Conversion Efficiency and Figure of Merit, CRC Handbook of Thermoelectrics, Boca Raton, FL, 1995.
[9] M. Cobble, in: D. Rowe (Ed.), Calculation of Generator Performance, CRC Handbook of Thermoelectrics, Boca Raton, FL, 1995.
[10] R.R. Heikes, R.W. Ure, in: R.R. Heikes, R.W. Ure (Eds.), Theoretical Calculation of Device Performance, Thermoelectricity: Science and Engineering, Interscience, New York, 1961.
[11] R. Bjørk, The universal influence of contact resistance on the efficiency of a thermoelectric generator, J. Electron. Mater. 44 (2015) 2869–2876.
[12] J. D'Angelo, E.D. Case, N. Matchanov, C.-I. Wu, T.P. Hogan, J. Barnard, C. Cauchy, T. Hendricks, M.G. Kanatzidis, Electrical, thermal, and mechanical characterization of novel segmented-leg thermoelectric modules, J. Electron. Mater. 40 (2011) 2051–2062.
[13] R. Li, B. Qiao, H. Shang, J. Zhang, C. Jiang, W. Zhang, Multi-component AlCrTaTiZrMo-nitride film with high diffusion resistance in copper metallization, J. Alloy. Comp. 748 (2018) 258–264.
[14] X. Zhu, L. Cao, W. Zhu, Y. Deng, Enhanced interfacial adhesion and thermal stability in bismuth telluride/nickel/copper multilayer films with low electrical contact resistance, Adv. Mater. Interfaces 5 (2018) 1801279.
[15] Dong-Jin Kim, Young-Bae Jung, Myoung-Bok Lee, Yong-Hyun Lee, Jong-Hyun Lee, J.-H. Lee, Applicability of ALE TiN films as Cu Si diffusion barriers, Thin Solid Films 372 (2000) 276.
[16] J.S. Becker, R.G. Gordon, Diffusion barrier properties of tungsten nitride films grown by atomic layer deposition from bis(tertbutylimido)bis(dimethylamido)tungsten and ammonia, Appl. Phys. Lett. 82 (2003) 2239.
[17] S. Rawal, D.P. Norton, KeeChan Kim, T.J. Anderson, L. McElwee-White, Ge/HfNx diffusion barrier for Cu metallization on Si, Appl. Phys. Lett. 89 (2006) 231914.
[18] C.-H. Wang, H.-C. Hsieh, H.-Y. Lee, A.T. Wu, Co-P diffusion barrier for p-Bi2Te3 thermoelectric material, J. Electron. Mater. 48 (2018) 53–57.
[19] P.Y. Chien, C.H. Yeh, H.H. Hsu, A.T. Wu, Polarity effect in a Sn3Ag0.5Cu/bismuth telluride thermoelectric system, J. Electron. Mater. 43 (2013) 284–289.
[20] C.-Y. Ko, A.T. Wu, Evaluation of diffusion barrier between pure Sn and Te, J. Electron. Mater. 41 (2012) 3320–3324.
[21] O.D. Iyore, T.H. Lee, R.P. Gupta, J.B. White, H.N. Alshareef, M.J. Kim, B.E. Gnade, Interface characterization of nickel contacts to bulk bismuth tellurium selenide, Surf. Interface Anal. 41 (2009) 440–444.
[22] T. Sugahara, Y. Ekubaru, N.V. Nong, N. Kagami, K. Ohata, L.T. Hung, M. Okajima, S. Nambu, K. Suganuma, Fabrication with semiconductor packaging technologies and characterization of a large-scale flexible thermoelectric module, Adv. Mater. Technol. (2018) 1800556.
[23] M. Edwards, K. Brinkfeldt, U. Rusche, T. Bukes, G. Gaiser, M. Da Silva, D. Andersson, The shear strength of nano-Ag sintered joints and the use of Ag interconnects in the design and manufacture of SiGe-based thermo-electric modules, Microelectron. Reliab. 55 (2015) 722–732.
[24] D. Crane, J. LaGrandeur, V. Jovovic, M. Ranalli, M. Adldinger, E. Poliquin, J. Dean, D. Kossakovski, B. Mazar, C. Maranville, TEG on-vehicle performance and model validation and what it means for further TEG development, J. Electron. Mater. 42 (2012) 1582–1591.
[25] C. Chen, S. Noh, H. Zhang, C. Choe, J. Jiu, S. Nagao, K. Suganuma, Bonding technology based on solid porous Ag for large area chips, Scr. Mater. 146 (2018) 123–127.
[26] C. Chen, S. Nagao, K. Suganuma, J. Jiu, T. Sugahara, H. Zhang, T. Iwashige, K. Sugiura, K. Tsuruta, Macroscale and microscale fracture toughness of microporous sintered Ag for applications in power electronic devices, Acta Mater. 129 (2017) 41–51.
[27] T. Fan, H. Zhang, P. Shang, C. Li, C. Chen, J. Wang, Z. Liu, H. Zhang, K. Suganuma, Effect of electroplated Au layer on bonding performance of Ag pastes, J. Alloy. Comp. 731 (2018) 1280–1287.
[28] C. Pei, C. Chen, K. Suganuma, G. Fu, Thermal stability of silver paste sintering on coated copper and aluminum substrates, J. Electron. Mater. 47 (2017) 811–819.